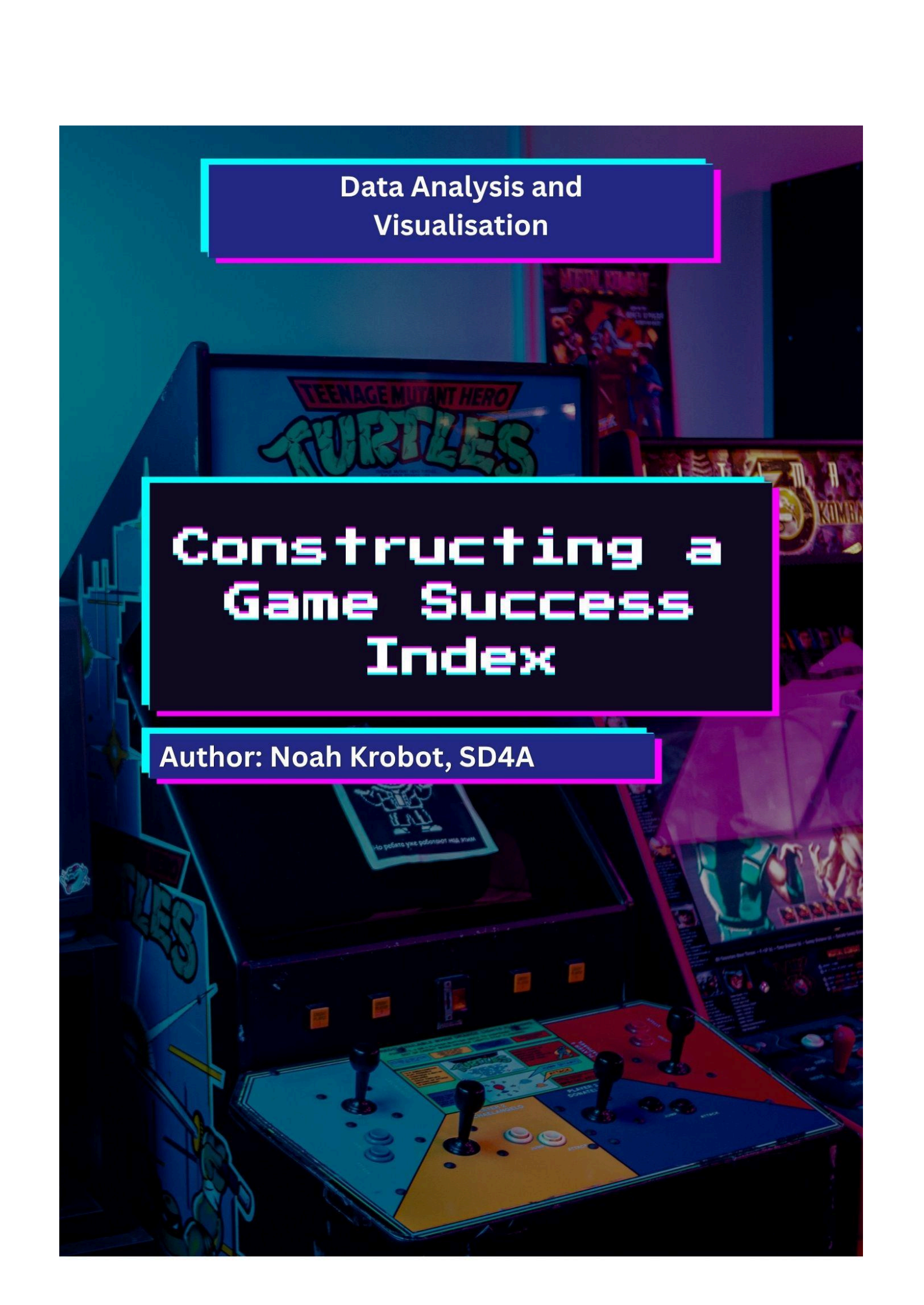




Data Analysis and
Visualisation

Constructing a Game Success Index

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0. Table of Contents

0. Table of Contents	2
1. Theoretical Framework	3
1.1. Research Question	3
1.2. Index Structure	3
2. Data Selection	4
2.1. Primary Dataset: Video Game Sales and Ratings	4
2.1.1. Strengths	5
2.1.2. Weaknesses	5
2.2. Secondary Dataset: Steam Video Games	5
2.2.1. Strengths	6
2.2.2. Weaknesses	6
2.3. Merging Datasets	6
3. Imputation of Missing Data	7
3.1. Player reviews and scores	7
3.1.1. Rounding the steam scores	7
3.1.2. Single player column	7
3.2. Critic reviews	8
3.3. Removing duplicates	8
3.4. Final columns kept	8
3.5. Check numeric columns or missing data	9
3.6. Detecting and capping outliers	9
4. Multivariate analysis	10
4.1. Correlation Heatmap	10
4.2. Principal Component Analysis	11
4.3. K-means Clustering	11
4.3.1. Verifying the right k value was found	11
4.3.2. Emerged clusters	12
4.4. MLR	14
5. Normalization	14
5.1. Dealing with Skewed data:	14
5.2. Min-Max normalisation	15
Originally, dealing with outliers was meant to be done in the normalisation step, especially because Min-Max Scaler requires outliers to be cleared up. However, the outliers were already dealt with just before the Multivariate Analysis step.	15
6. Weighting and Aggregation	15
6.1. Calculating Sub-indices	15
6.2. Calculating Index	15
7. Final Results	16
7.1. Top Games	16
7.2. Average Sub-index by Genre	17
8. Link to other Indices	18
8.1. Top 10 IMDb games compared to Game Success index	18
8.2. Top 10 The Punished Backlog games compared to Game Success index	19
8.3. Top 10 OpenCritic games compared to Game Success index	20
8.4. Comparing external indices with each other	21
9. Version Control	21

1. Theoretical Framework

1.1. Research Question

According to Georgia Entertainment, Sherwood and MediaCat UK, the video game industry generates revenue far greater than the movie and music industry combined. Therefore, it is extremely important that video game publishers take a piece of that revenue, in the most successful way possible. To achieve that, publishers must adapt to the user needs, wants and expectations. If you open a Metacritics or a Steam page, and pick a random game, you'll find yourself looking at the audience score almost immediately. The scores construct a reputation for the game, which might influence the game's further commercial success or failure.

The goal of this project is to construct a Video Game Success Index - an indicator to compare and display relationships between commercial success within regional or global markets, and audience reviews across different video game platforms. The problem question is:

“What is the relationship between video game sales and the critical and player feedback a title receives?”

Specifically, the index tends to answer whether video game publishers who receive positive audience feedback achieve bigger success. The answers may help small indie developers and all the way to big video game corporations.

1.2. Index Structure

The Video Game Success Index is formed using three different sub-indexes, each capturing different aspects of video game success. The sub-indexes are:

- Sales sub-index: This sub-index covers video game commercial performance across North America, the European Union, Japan and the rest of the markets. Summing up all markets, we get global sales, which are also part of the subindex.
- Critic score sub-index: This sub-index covers opinions written by video game critics, in this case sourced by Metacritic and VGChartz. It combines the normalised critic score with critic count. The critic count is included to address disproportion of the data. Some video games might have only one critic count, while some might have a thousand more. Therefore, the game with one critic score being 9/10 is much less trustworthy than a game with a thousand critics with 9/10 as well.
- Player Score sub-index: This sub-index covers player feedback by combining Metacritic user reviews with Steam game reviews. It also includes player score count, to prevent untrustworthiness amongst the data, the same as the previous sub-index.

Here is an example of what the final dataset might look like at the end:

Name	Platform	Sales SubIndex	Critic_SubIndex	Player_SubIndex	Game Success Index
Grand Theft Auto V	PC	Number	Number	Number	Number
The Witcher III	PS4	Number	Number	Number	Number
Cyberpunk 2077	PC	Number	Number	Number	Number

2. Data Selection

2.1. Primary Dataset: Video Game Sales and Ratings

This dataset sources information from VHChartz and Metacritic. It provides a broad regional and global sales record of seventeen thousand video games released across major platforms (e.g. PlayStation 4, Xbox 360, Nintendo Wii), and their critic and player review scores.

The key columns selected from this dataset are:

Column	Type	Description
Name	Categorical	Game's title.
Platform	Categorical	Gaming platform (e.g. Nintendo Wii, PC).
Genre	Categorical	Game's genre, used to group games.
NA_Sales	Continuous	Sales in North America in millions.
EU_Sales	Continuous	Sales in the European Union in millions.
JP_Sales	Continuous	Sales in Japan are in the millions.
Other_Sales	Continuous	Sales in the rest of the world in millions.
Global_Sales	Continuous	Total sales in millions.
Critic_Score	Continuous	Metacritic score from 0 to 100.

Critic_Count	Discrete	Number of critics reviewing the game.
User_Score	Continuous	Metacritic user (player) reviews scores from 0 to 10.
User_Count	Discrete	Number of users (players) reviewing the game.

Some of the columns were simply not needed, and were therefore excluded: Publisher and Year_of_release to reduce noise, Rating column as it doesn't relate to game score, but just limits the audience.

2.1.1. Strengths

- Broad dataset: the dataset contains around 17 000 games across 31 platforms.
- Regional sales splits with no missing values: the datasets present sales columns really well as it gives sales numbers broken down across multiple regions.
- Scores taken from multiple sources (VGChartz and Metacritic)

2.1.2. Weaknesses

- Steam not included: the dataset does not include one of the largest PC gaming platforms, missing out on a vast number of games.
- Missing scores: a lot of rows simply miss values in critic_score, critic_count, user_score and user_count.
- Slightly outdated data: the dataset covers only games up to 2017. Therefore, games published in the last 7 years aren't covered.

2.2. Secondary Dataset: Steam Video Games

This dataset sources video game data from PC gaming platform Steam. It contains around 200 000 user interactions split between 27 000 video games.

The key columns selected from this dataset are:

Column	Type	Description
name	Categorical	Game's title.
positive_ratings	Discrete	Number of positive steam reviews.
negative_ratings	Discrete	Number of negative steam reviews.
genres	Categorical	Game's genre.

This dataset is added to apply a depth to the primary dataset. It provides a lot of data regarding gaming on PC, which is absolutely crucial to cover.

Some of the columns were simply not needed to solve the problem question. The next columns were removed to reduce noise and unnecessary data overload: appid, release_date, english, developer, publisher, categories, steamspy_tags, achievements, average_playtime, median_playtime, owners, and price. The platform column was also removed. Even though Platform column is used in dataset one as it splits games amongst different consoles and devices; platform column in this case simply splits the games based on operating systems: Windows, Mac, Linux. All of these operating systems fall into the dataset 1's PC platform.

2.2.1. Strengths

- Broad dataset: the dataset contains around 27 000 games.
- No missing data

2.2.2. Weaknesses

- Strictly PC: the dataset shows games available only at the PC version of Steam. That wouldn't be enough to solve the problem question, and therefore cannot be a standalone dataset in this case.

2.3. Merging Datasets

Datasets were merged with left joining Steam dataset onto Video Game dataset and by creating matching keys using Name columns in both datasets. Matching key columns were stripped of whitespaces and transformed to lowercase letters. However, as Steam doesn't have a single user score as the Video Game dataset does, I've turned Steam's positive and negative ratings into a steam_positive_review_ratio column, by dividing positive_ratings by a sum of positive_ratings and negative_ratings. Later, steam_positive_review_ratio is transformed to simply steam_score.

After the merge, there were **17426** rows left.

3.Imputation of Missing Data

3.1. Player reviews and scores

3.1.1. Rounding the steam scores

Firstly, I rounded the steam score to 1 decimal place and scale it up so its values are between 0 and 10. The reason for that is that user_score uses metacritic user (player) reviews scores from 0 to 10 (e.g. 6.1), while the steam_scores was e.g. 0.611 instead of 6.1.

3.1.2. Single player column

I had to create a single player score column (later called **transformed_user_score**) from two sources: primary dataset (User_Score, User_Count) and secondary database (steam_score, steam_total_reviews).

Calculations

The first problem was scale disparity between User_count and steam_total_reviews. The first gave a count in the range of a few hundreds to a few thousand, the largest being 8 715. The latter however commonly reach a number of 35 thousands and hundreds of thousands.

To use both columns correctly, I had to calculate a log weighted average. I took a natural log of steam_total_reviews and User_Count to put them into a single comparable range. After that, I calculated the weighted average.

Scenarios

After the merge, some rows had only User_Score, some only steam_scores, some had both and some had neither, giving me four possible scenarios.

- Scenario 1 (only User_Score): Populates **transformed_user_score** column with **only user_score**.
- Scenario 2 (only steam_score): Populates **transformed_user_score** column with **only steam_score**.
- Scenario 3 (both User_Score and steam_score): For this, calculating the weighted average was necessary.
- Scenario 4 (no User_Score and steam_score): Rows like this were removed from the dataset entirely. Due to a large quantity of such rows, imputing the player scores with medians would result in a high level of inaccuracy .

3.2. Critic reviews

Missing values in Critic_Score and Critic_Count were imputed by calculating the median within the genre that video games fall into. This was done to prevent a high level of inaccuracy within the dataset, and to prepare the dataset to be as precise as possible. For

example, a game that falls under the ‘Adventure’ genre will not get the same critical score and count as a game that falls under ‘Sports’. By doing this, I’ve

3.3. Removing duplicates

Some rows had equal game names, but different platforms (e.g. GTA V on Xbox, and on PS4). These rows were left in, as they will give a more in-depth view of on which platform a certain game is more successful. For example, instead of saying GTA V is commercially successful, we can say GTA V is commercially successful on Xbox more than on PS4.

However, some rows had both the same name and the platform set as the same. These rows were considered **duplicates** and therefore had to be removed, but the first occurrence of the duplicates were kept in the dataset.

3.4. Final columns kept

The remaining columns were: Name, Platform, Genre, NA_Sales, EU_Sales, JP_Sales, Other_Sales, Critic_Score, Critic_Count, transformed_user_score

The removed columns at this step were:

Variable	Reason for removal
Publisher	not important in this scenario
Year of Release	not important in this scenario
User Score	transformed into transformed_user_score
User Count	transformed into transformed_user_score
Global Sales	redundancy, as is just a sum of all sales columns

3.5. Check numeric columns or missing data

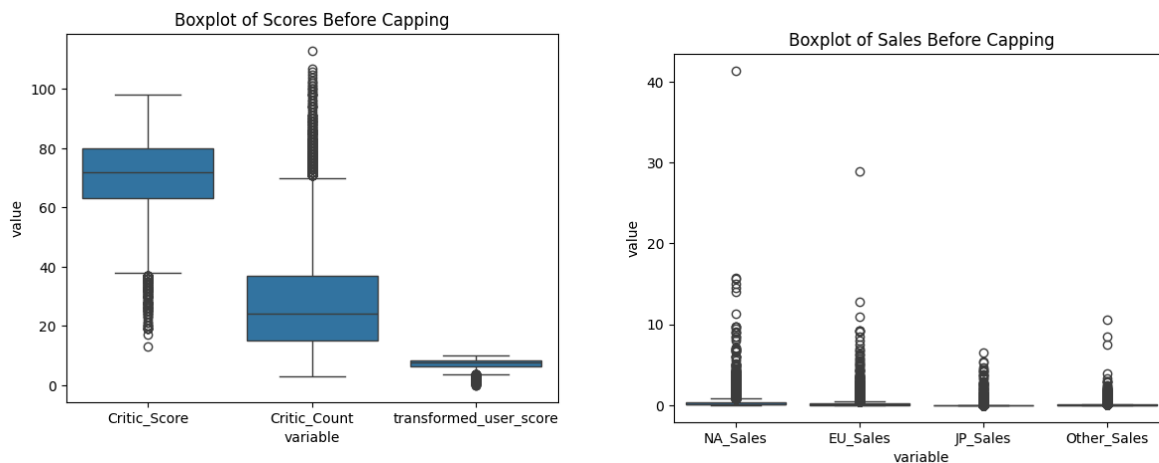
For the problem question to be solved properly, all numeric columns had to be 100% numeric. All of the next numeric columns were checked for any non-numeric values: 'NA_Sales', 'EU_Sales', 'JP_Sales', 'Other_Sales', 'Global_Sales', 'Critic_Score', 'Critic_Count', 'transformed_user_score'.

Besides that, all rows were checked for any missing values, and there were none.

3.6. Detecting and capping outliers

It is crucial to deal with outliers before correlation heatmap, PCA and K-Means Clustering. Keeping them in the data would result in incorrect multicollinearity levels, skewing in the PCA and incorrect clustering.

The next boxplots show that outliers are present in both score and especially in sales columns:



Removing the outliers would remove games that are extremely popular and critically acclaimed (e.g. GTA V), or would remove the not critically acclaimed games (e.g. Angry Birds Star Wars). To handle this better, outliers were capped at the 5th and 95th percentiles.

The results confirm the capping was successful. All four sales columns had around 400 outliers before capping, and none after. The three score columns had from 692 to 754 outliers, and none after capping.

4. Multivariate analysis

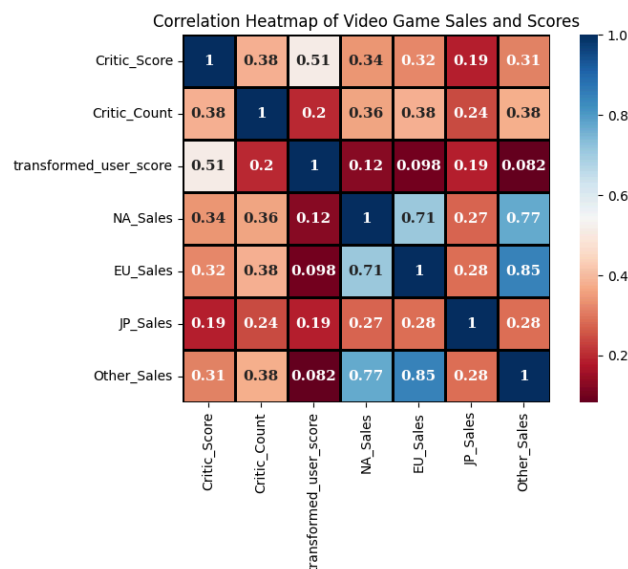
4.1. Correlation Heatmap

Prior to applying PCA, a correlation heatmap was produced to assess linear relationships between variables. PCA assumes linearity, and the heatmap confirms that meaningful linear relationships exist across the numeric variables, justifying its use.

However, the presence of sales outliers may weaken the linearity assumption, which is noted as a limitation.

Before doing the PCA and clustering, a correlation heatmap was made to observe the linear relationships between the variables. It is crucial to do this before PCA, as PCA only works on linear data.

Here is the produced heatmap:



The heatmap states that all sales besides JP_Sales are moderately or strongly correlated with each other. This was expected as games that would sell well in the EU will likely sell well in North America. However, Japan's video game market is slightly different - there's more original titles, and more titles from China and Korea pushed onto the market. Therefore, it makes sense that JP_Sales isn't correlated with the rest.

On the other hand, score variables (Critic_score, Critic_count and transformed_user_score) aren't strongly or moderately correlated to each other, which is a great sign that the database captures different information.

4.2. Principal Component Analysis

PCA was done to verify that sales columns and score columns give different information about the data, and besides that to detect which variables were the best within the dataset. The best PCs for that were chosen using Kaiser Criterion, leaving only PC1 and PC2.

In the results, it is noticeable that PC1 was dominated by sales columns making it a commercial dimension. However, PC2 was dominated by score columns making it a quality

dimension. This proves that sales and scores are not redundant and describe different aspects of a game's success. Ultimately, both can be included into the Game Success Index.

The top five variables that carry the most information within PCs are transformed_user_score, Critic_Score, Other_Sales, EU_Sales, and NA_Sales. This leveling helps with constructing sub-indices later.

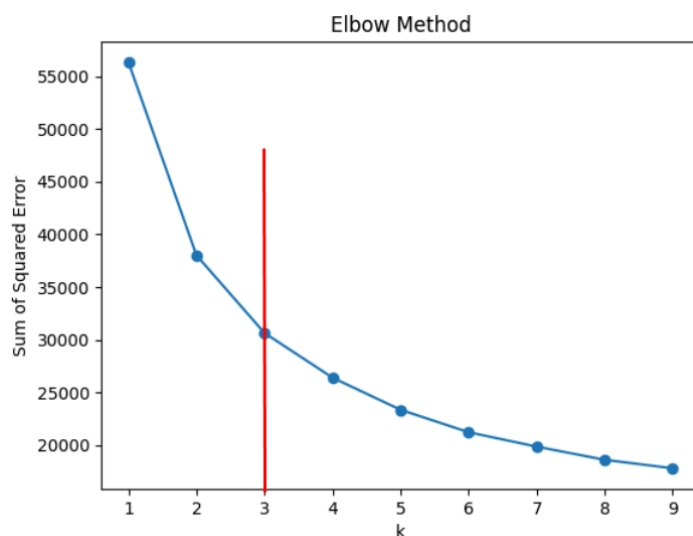
4.3. K-means Clustering

To explore any natural groupings of video games within the dataset K-means Clustering was applied. The groups would observe video games from both commercial and quality viewpoints.

4.3.1. Verifying the right k value was found

The elbow method was implemented, and the number 3 again points to a place where the graph changes significantly after.

After 3, the differences in each point are less significant, confirming that 3 is the correct choice for k value. The red line placed at $k = 3$ on the makes it easier to notice that border.



4.3.2. Emerged clusters

After the clustering step, three different clusters appeared. Each tells a unique story about the relationship between sales columns and score columns.

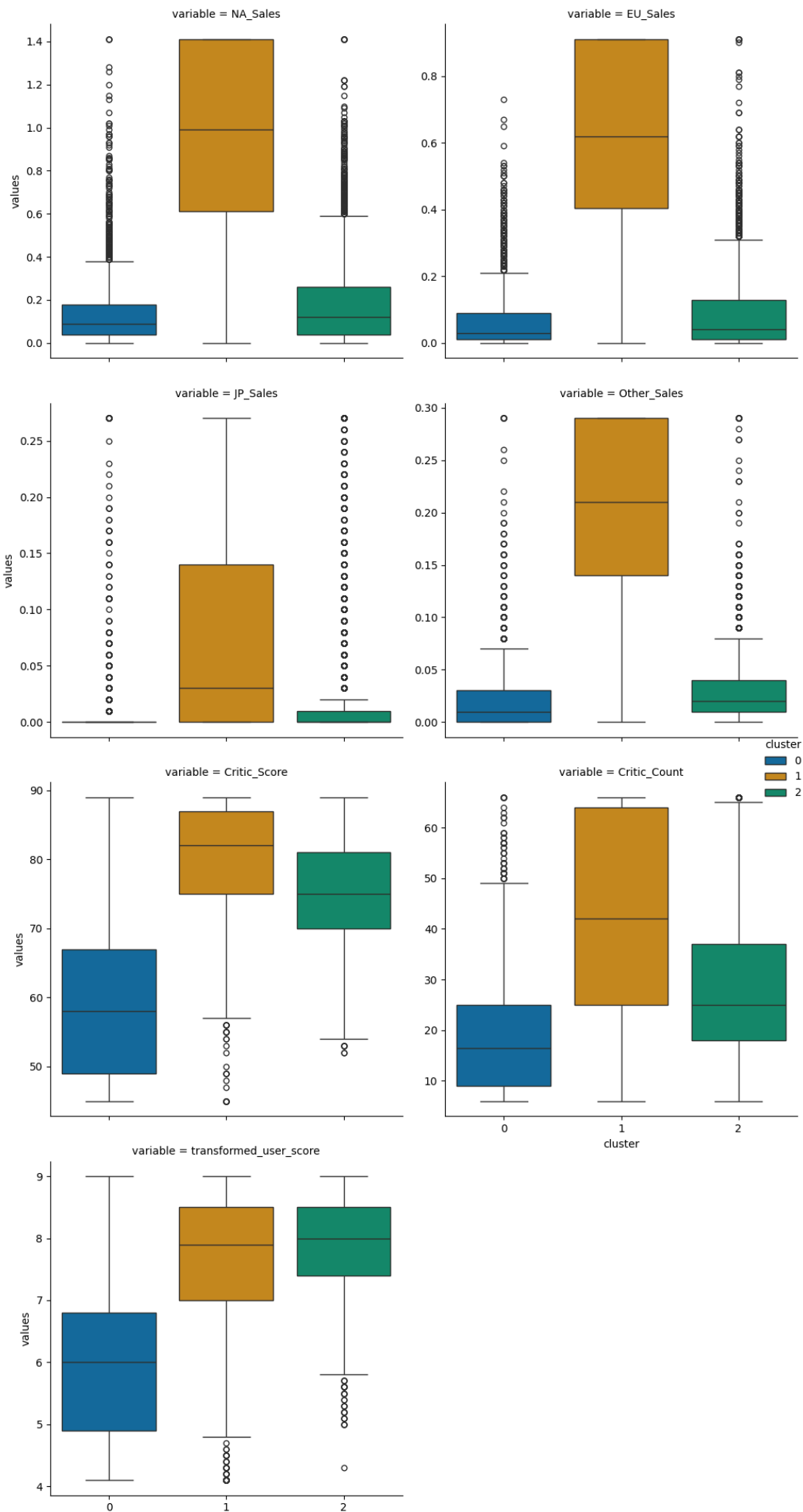
Cluster 0 is a cluster of low commercial performance and average or below average scores. Out of a total of 8047 video games, 2535 (32% of video games) fall under this category. This proves that it can be hard to make a game of good quality, and make it a global success.

Cluster 1 is a cluster of high commercial performance and strong scores. Out of a total of 8047 video games, 1213 fall under this category (15% of video games). This proves that the

critical and player scores can take the game a long way in terms of commercial success, but this is rare.

Cluster 2 is a cluster of poor commercial performance and strong or moderate scores. Out of a total of 8047 video games, 4299 fall under this category (53% of video games). This proves that the majority of well-reviewed games aren't a big commercial hit, and ultimately proves that the quality alone does not guarantee commercial success.

The three clusters point out that the video game market is disproportionate, and it would be wise to observe other factors besides commercial and scores. Variables like: timing, investments, publisher worth, inflation and others might be good to consider.



4.4. MLR

Multiple Linear Regression was considered but not performed. In this scenario, MLR would be redundant, and would not make a significant contribution, as PCA and Clustering already have a clear picture of the relationships within the data.

5. Normalization

5.1. Dealing with Skewed data:

According to SmartPLS, Skewness measures the symmetry of a variable's distribution. If the distribution stretches toward the right or left tail, it's skewed. Negative skewness indicates larger values, while positive skewness indicates smaller values. A skewness value between -1 and +1 is excellent, while -2 to +2 is acceptable. Values beyond -2 and +2 should be removed.

The results of skewness check:

Variable	Skewness value	Interpretation
Critic_Score	-0.53	PASS
Critic_Count	0.82	PASS
transformed_user_score	-0.79	PASS
NA_Sales	1.85	PASS
EU_Sales	2.01	FAIL
JP_Sales	2.62	FAIL
Other_Sales	1.93	PASS

Considering two variables are highly skewed, log transformation was applied to the dataset. The transformation did fix the EU_Sales column, making it acceptable, but it didn't fix JP_Sales. The reason for that is: JP_Sales has 5814 out of 8047 values set to 0 (72% of the column), as the Japanese market usually has different game availability and reach than EU and North America and a lot of games simply did not hit the Japanese market.

5.2. Min-Max normalisation

The Min-Max Scaler will produce easy-to-interpret data, scaled from 0 to 1, and it'll be easy to interpret the results in the index and sub-indices. Another scaler that was considered was the Standard Scaler. However, it can give more confusing and harder-to-interpret results with its negative values. Therefore, it was quickly dropped.

Originally, dealing with outliers was meant to be done in the normalisation step, especially because Min-Max Scaler requires outliers to be cleared up. However, the outliers were already dealt with just before the Multivariate Analysis step.

6. Weighting and Aggregation

6.1. Calculating Sub-indices

The first sub-index created is Sales_SubIndex. The sub-index was calculated by getting a mean value of four sales columns: NA_Sales, EU_Sales, JP_Sales, Other_Sales.

The second sub-index created is Critic_SubIndex. In PC1, critic_count and Critic_score are almost the same. However, in PC2, the Critic_Score is 3 times bigger than Critic_Count. Therefore, the Critic_SubIndex is a product of 75% Critic_Score, and 25% Critic_Count.

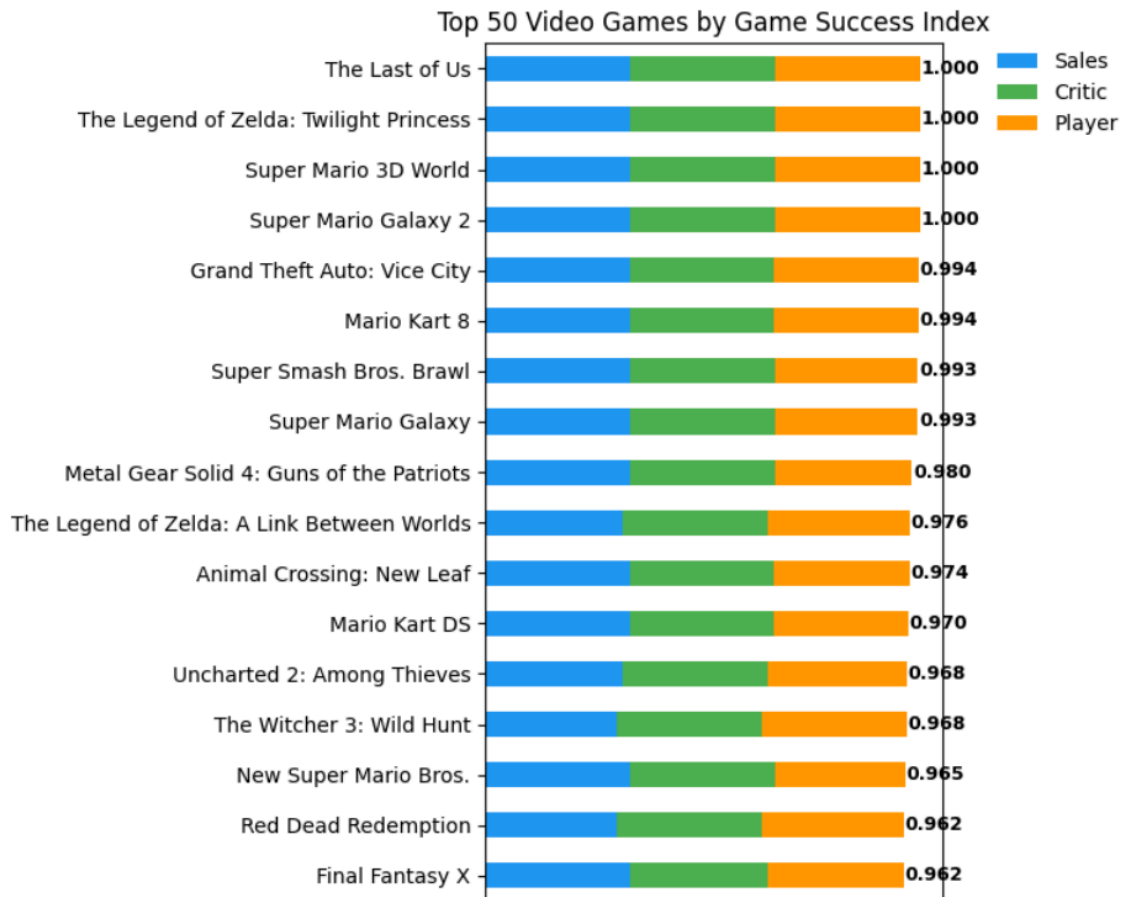
The third and last sub-index created is Player_SubIndex, which equals the transformed_user_score already created at the Imputation of Missing Data step.

6.2. Calculating Index

Finally, the main index (Game Success Index) is just an average of all three subindexes. They all carry the same weight of 33%, as they can be considered to have the same effect on the data. For example, even though PC1 was dominated by sales columns, that was presumably because there's four different sales columns. PC2 was fully dominated by score columns. With taking both points into account, it makes sense to give every sub-index equal amount or worth.

7. Final Results

7.1. Top Games

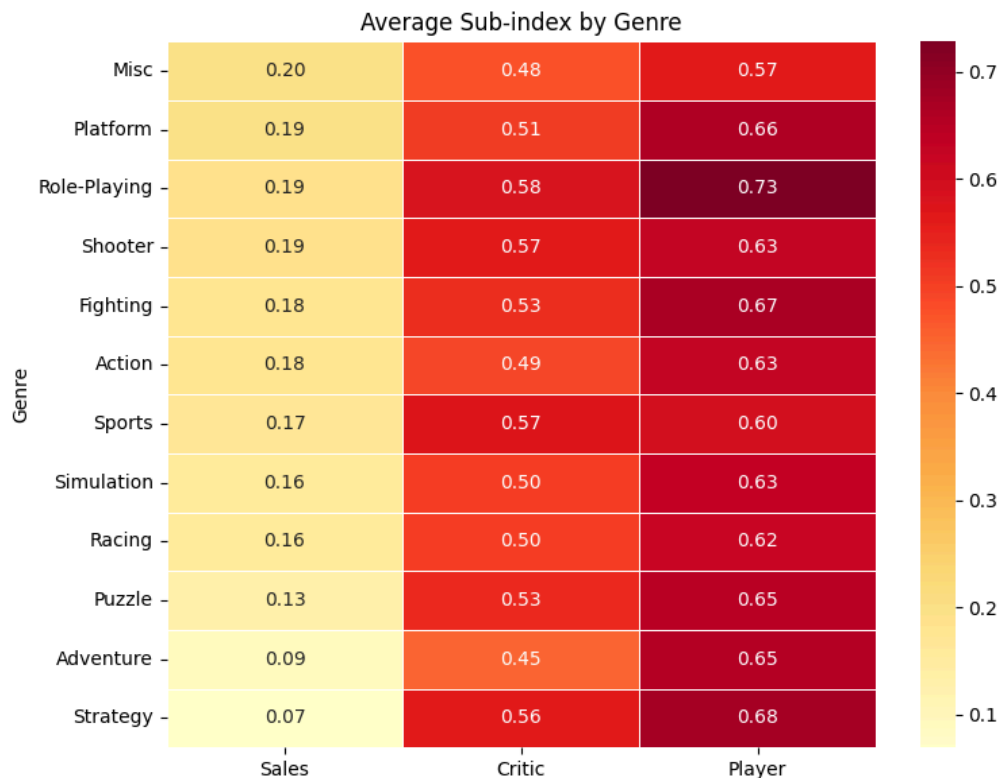


In the final results chart, it is clearly seen that the top four games (The Last of Us, Legend of Zelda: Twilight Princess, Super Mario 3D World, Super Mario Galaxy 2) all score a perfect 1.000 score. This means they had the biggest possible values for all three sub-indices.

However, other game's bars vary. Some games like Red Dead Redemption or The Witcher 3: Wild Hunt have a smaller Sales (blue colored) segment, but still rank high, as they both have a strong Critic and Player scores.

This supports the earlier clustering finding that good reviews alone don't guarantee sales, but they can still produce a high index score.

7.2. Average Sub-index by Genre



The heatmap presents a consistent pattern across all video game genres.

Sales sub-index values are much lower than both Critic and Player sub-indices.

Not a single genre exceeds 0.20 in Sales. However, the Critic are between 0.45 and 0.58, and Player scores from 0.57 to 0.73. Not a single genre jumps out as an extreme outlier proving that it is the best shot at achieving commercial success.

This supports the earlier clustering finding that commercial breakthrough is rare regardless of critic and player scores, or of the genre of the game.

This is extremely important for all game developers (from indie to big corporations):

No genre is a guarantee of a consistent commercial success.

8. Link to other Indices

8.1. Top 10 IMDb games compared to Game Success index

Number	Game Success Index	IMDb
1	The Last of Us	The Last of Us
2	The Legend of Zelda: Twilight Princess	The Witcher 3: Wild Hunt
3	Super Mario 3D World	The Legend of Zelda: Ocarina of Time
4	Super Mario Galaxy 2	Final Fantasy VII
5	Grand Theft Auto: Vice City	Mass Effect 2
6	Mario Kart 8	Metal Gear Solid
7	Super Smash Bros. Brawl	Metal Gear Solid 3: Snake Eater
8	Super Mario Galaxy	Star Wars: Knights of the Old Republic
9	Metal Gear Solid 4: Guns of the Patriots	Batman: Arkham City
10	The Legend of Zelda: A Link Between Worlds	Chrono Trigger

Full match(green): The two lists share only one game - The Last of Us.

Partial matches (orange): Both lists include The Legend of Zelda games, but different chapters of the same series. Also, both lists include a Metal Gear Solid chapter. Therefore, it would be wise to dive deeper in the future which makes The Legend of Zelda and Metal Gear Solid games so highly ranked.

IMDb's list includes some games like The Witcher 3: Wild Hunt, Final Fantasy VII and Mass Effect 2, which do not appear in the Game Success Index at all.

Finally, the two lists have a matching percentage of 10% (1 matching game), and partial matching of 25 % .

8.2. Top 10 The Punished Backlog games compared to Game Success index

Number	Game Success Index	The Punished Backlog
1	The Last of Us	The Legend of Zelda: Ocarina of Time
2	The Legend of Zelda: Twilight Princess	Mass Effect 2
3	Super Mario 3D World	The Witcher 3: Wild Hunt
4	Super Mario Galaxy 2	Super Smash Bros. Melee
5	Grand Theft Auto: Vice City	Stardew Valley
6	Mario Kart 8	The Legend of Zelda: Breath of the Wild
7	Super Smash Bros. Brawl	The Legend of Zelda: Majora's Mask
8	Super Mario Galaxy	The Elder Scrolls V: Skyrim
9	Metal Gear Solid 4: Guns of the Patriots	The Legend of Zelda: Twilight Princess
10	The Legend of Zelda: A Link Between Worlds	Bloodborne

Full match (green): The two lists have only one matching game - The Legend of Zelda: Twilight Princess.

Partial matches (orange): Both lists include The Legend of Zelda games, but different chapters of the same series. Also, both lists include a Super Smash Bros. chapter. Therefore, it would be wise to analyse Super Smash Bros. games in detail in the future. Besides that, it becomes even more clear that The Legend of Zelda games would be very interesting to observe in detail, as they appear in the second comparison as well.

IMDb's list includes some games like Mass Effect 2, The Witcher 3: Wild Hunt and Stardew Valley, which do not appear in the Game Success Index at all.

Finally, the two lists have a matching percentage of 10% (1 matching game), and partial matching of 30 % .

8.3. Top 10 OpenCritic games compared to Game Success index

Number	Game Success Index	OpenCritic
1	The Last of Us	Super Mario Odyssey
2	The Legend of Zelda: Twilight Princess	The Legend of Zelda: Breath of the Wild
3	Super Mario 3D World	Grand Theft Auto IV
4	Super Mario Galaxy 2	The Last of Us
5	Grand Theft Auto: Vice City	Persona 5
6	Mario Kart 8	Divinity: Original Sin 2
7	Super Smash Bros. Brawl	Journey
8	Super Mario Galaxy	The House in Fata Morgana
9	Metal Gear Solid 4: Guns of the Patriots	Undertale
10	The Legend of Zelda: A Link Between Worlds	Super Mario 3D World

Full match(green): The two lists share two games - The Last of Us and Super Mario 3D World. As The Last of Us appears again as a matching game, it proves that it would be interesting to analyse the reason behind that in more detail in future.

Partial matches (orange): Both lists include The Legend of Zelda games, but different chapters of the same series. Also, both lists include Super Mario and Grand Theft Auto chapters. Therefore, it would be wise to analyse Super Mario and Grand Theft Auto games in detail in the future. Besides that, it becomes even more clear that The Legend of Zelda games would be very interesting to observe in detail, as they appear in all comparisons so far.

IMDb's list includes some games like Persona 5, Divinity: Original Sin 2 and Journey, which do not appear in the Game Success Index at all.

Finally, the two lists have a matching percentage of 10% (1 matching game), and partial matching of 40 % .

8.4. Comparing external indices with each other

Number	Game Success Index	IMDb	The Punished Backlog	OpenCritic
1	The Last of Us	The Last of Us	The Legend of Zelda: Ocarina of Time	Super Mario Odyssey
2	The Legend of Zelda: Twilight Princess	The Witcher 3: Wild Hunt	Mass Effect 2	The Legend of Zelda: Breath of the Wild
3	Super Mario 3D World	The Legend of Zelda: Ocarina of Time	The Witcher 3: Wild Hunt	Grand Theft Auto IV
4	Super Mario Galaxy 2	Final Fantasy VII	Super Smash Bros. Melee	The Last of Us
5	Grand Theft Auto: Vice City	Mass Effect 2	Stardew Valley	Persona 5
6	Mario Kart 8	Metal Gear Solid	The Legend of Zelda: Breath of the Wild	Divinity: Original Sin 2
7	Super Smash Bros. Brawl	Metal Gear Solid 3: Snake Eater	The Legend of Zelda: Majora's Mask	Journey
8	Super Mario Galaxy	Star Wars: Knights of the Old Republic	The Elder Scrolls V: Skyrim	The House in Fata Morgana
9	Metal Gear Solid 4: Guns of the Patriots	Batman: Arkham City	The Legend of Zelda: Twilight Princess	Undertale
10	The Legend of Zelda: A Link Between Worlds	Chrono Trigger	Bloodborne	Super Mario 3D World
Matching percentage	30 %	40 %	50 %	30 %

When compared in matching with other indices, the Game Success Index and OpenCritic are the lowest ranking ones. The matching percentage varies in range from 30 % to 50%.

Therefore, it is clear that no indices should be considered wrong or bad. Presumably, the indices were looking at different factors when producing a top 10 list, proving that the Game Success Index is a trustworthy source, at least when looking at top 10 games.

9. Version Control

The GitHub link to the code repository behind constructing the Game Success Index is here: https://github.com/NoahKrobot/CA1_Video_Game_DAV.

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